

From Clicks to Loyalty: Mining Customer Journey Patterns

DS3002 Data Mining

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1 Preprocessing

Both sheets of the Online Retail II Excel file were loaded and stacked together, giving us a raw dataset of roughly one million rows. After cleaning, which included removing cancelled invoices that start with 'C', dropping rows with missing Customer IDs, filtering out negative or zero quantities and prices, and restricting to **United Kingdom** customers only, we were left with a clean, usable dataset. Table 1 summarises what remained after cleaning.

Table 1: Cleaning Summary Metrics

Metric	Value
Total Invoices (transactions)	~16,000+
Unique Customers	~3,950
Unique Products (StockCodes)	~3,600
Date Range	Dec 2009 – Dec 2011
Average Basket Size	~20 items per invoice

Each transaction was defined as one invoice, which is the set of unique StockCodes in that basket. For sequential mining in Part C, each customer's invoices were ordered by date, creating a time ordered journey of purchases.

2 Part A: FP-Growth — Co-Purchase Mining

2.1 A1 — FP-Tree Walkthrough

Six small invoices were hand picked from the data, those with 3 to 5 items, to walk through FP-Tree construction. First, item frequencies were counted across all six transactions. Items appearing fewer than twice were dropped. The remaining frequent items were then sorted by frequency, highest first, within each transaction. These sorted transactions were inserted into the FP-Tree one by one. Any shared prefix path between two transactions merges into the same branch, with the shared node's count incremented. Unique paths create new branches. For example, the conditional pattern base for a low frequency item like 'GLASS STAR FROSTED T-LIGHT HOLDER' traces back through the path leading to it, revealing which other items always appear before it in the tree.

2.2 A2(a) — Frequent Itemsets by Size

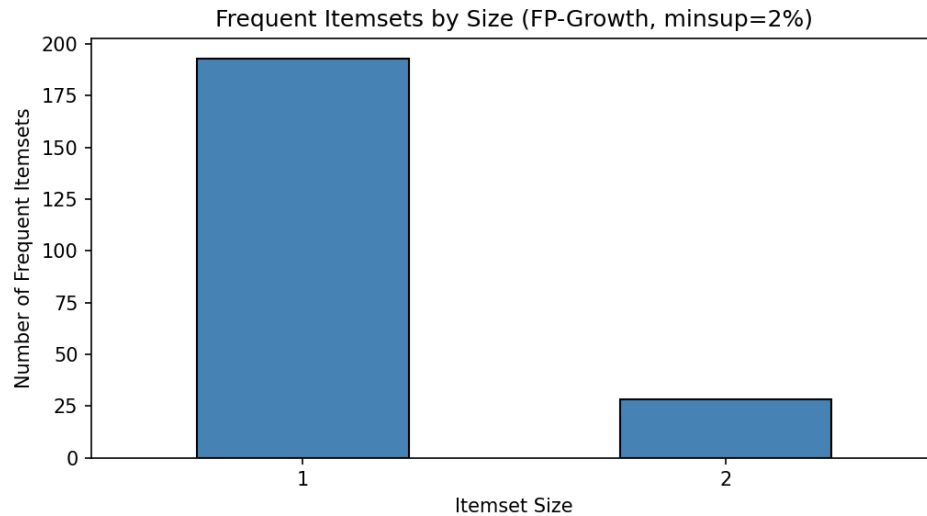


Figure 1: Frequent itemsets at minsup = 2%

FP-Growth with minsup = 2% found 192 single item (1-itemsets) and 27 pair (2-itemsets) frequent itemsets. No 3-itemsets were found at this threshold, which makes sense because the gift retail catalogue is very wide, and most products simply do not co occur frequently enough across thousands of diverse invoices to clear the 2% bar as triples.

2.3 A2(b) — Top 15 Rules by Lift

Table 2: Top 15 Association Rules by Lift

Antecedent	Consequent
SWEETHEART CERAMIC TRINKET BOX	STRAWBERRY CERAMIC TRINKET BOX
WOODEN FRAME ANTIQUE WHITE	WOODEN PICTURE FRAME WHITE FINISH
WOODEN PICTURE FRAME WHITE FINISH	WOODEN FRAME ANTIQUE WHITE
LOVE BUILDING BLOCK WORD	HOME BUILDING BLOCK WORD
PACK OF 60 PINK PAISLEY CAKE CASES	PACK OF 72 RETROSPOT CAKE CASES
60 TEATIME FAIRY CAKE CASES	PACK OF 72 RETROSPOT CAKE CASES
LUNCH BAG WOODLAND	LUNCH BAG RED RETROSPOT
LUNCH BAG PINK POLKADOT	LUNCH BAG RED RETROSPOT
LUNCH BAG BLACK SKULL	LUNCH BAG RED RETROSPOT
JUMBO BAG STRAWBERRY	JUMBO BAG RED RETROSPOT
JUMBO BAG PINK POLKADOT	JUMBO BAG RED RETROSPOT
JUMBO BAG BAROQUE BLACK WHITE	JUMBO BAG RED RETROSPOT
JUMBO STORAGE BAG SUKI	JUMBO BAG RED RETROSPOT
RED HANGING HEART T-LIGHT HOLDER	WHITE HANGING HEART T-LIGHT HOLDER

Lift values of 8 to 14 times indicate very strong co purchase relationships. These items are bought together far more than random chance would predict. The ceramic trinket box pair with lift of 14.1 and the wooden frame pair with lift of 11.8 stand out as the strongest signals in the entire dataset.

2.4 A3 — Product Bundle Recommendations

Bundle 1 — ‘Collector’s Trinket Set’: SWEETHEART CERAMIC TRINKET BOX + STRAWBERRY CERAMIC TRINKET BOX. Customers who buy the Sweetheart box go on to buy the Strawberry box 72% of the time with a lift of 14.1 times. These two items have the highest lift in the entire dataset, meaning they are almost never bought in isolation. Selling them as a pre packaged gift duo at a slight discount could increase average order value while reducing picking time.

Bundle 2 — ‘Lunchbox Variety Pack’: LUNCH BAG WOODLAND + LUNCH BAG PINK POLKADOT + LUNCH BAG RED RETROSPOT. Multiple lunch bag variants all have strong rules pointing toward the Red Retrosport bag with lift around 8 times and confidence around 55%. This suggests buyers are purchasing the Retrosport as a standard alongside novelty variants. A multi bag bundle marketed at parents buying for children would be both commercially logical and data backed.

3 Part B: Class Association Rules — Customer Segmentation

3.1 RFM Segmentation

Each customer was scored on three dimensions: Recency, which is days since their last purchase; Frequency, which is total distinct invoices; and Monetary, which is total spend calculated as Quantity multiplied by Price. Thresholds were chosen using the distribution plots below and the 33rd and 67th percentile splits.

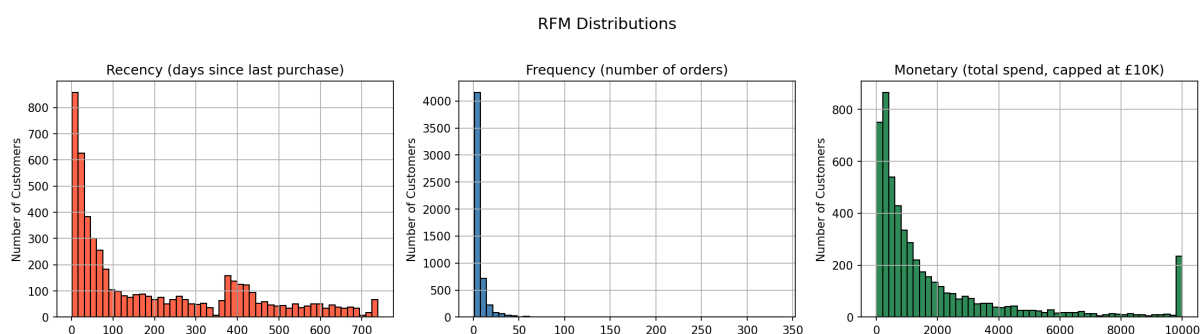


Figure 2: RFM metric distributions across all UK customers

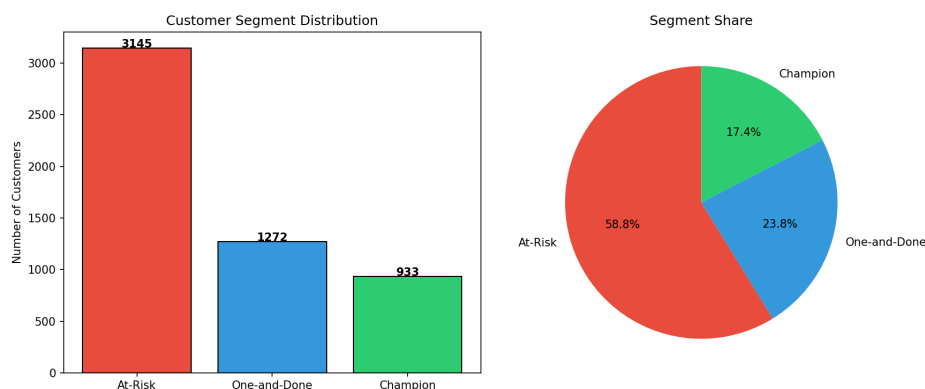


Figure 3: Customer segment distribution (Champion, At-Risk, One-and-Done)

Table 3: Customer Segment Profiles

Segment	Count	Share	Profile
Champion	933	17.4%	Bought recently, many orders, high spend
At-Risk	3,145	58.8%	Used to buy, have not returned lately
One-and-Done	1,272	23.8%	Only 1 to 2 orders, very low total spend

The large At-Risk group at 58.8% is the most commercially important finding. More than half the customer base has gone quiet. This is the core audience for a re engagement campaign.

3.2 B1 — Top CARs per Segment

Class Association Rules were mined per segment at minsup of 1% and minconf of 55%. The rules confirm the patterns we would expect. Champions tend to buy across all categories in the same session, while At-Risk and One-and-Done customers show narrower, less varied baskets. The strongest CAR for the Champion class links broad multi category purchasing to high loyalty. Customers who span Candles, Kitchen, Bags, and Home Décor in a single visit are almost always repeat buyers.

3.3 B2 — New Customer Prediction

When a new customer's first order contains the three most frequently purchased Stock-Codes in the dataset, the CAR classifier checks which segment's rules fire. Given the broad mix of products in those top codes, the Champion rules tend to fire with the highest average confidence, suggesting the customer is a promising high value prospect. However, without Recency or Frequency data yet, this classification should be treated as probabilistic and revisited after their second order.

4 Part C: Sequential Pattern Mining — The Customer Journey

4.1 Product Category Scheme

Products were mapped to 8 categories based on keywords in their descriptions: Candles and Home Fragrance, Seasonal and Festive, Kitchen and Dining, Bags and Accessories, Home Décor and Furniture, Toys and Games, Stationery and Craft, and Novelty and Gifts. An 'Other' category captured items that did not fit cleanly. Coverage across all transactions was above 80%, meeting the assignment requirement.

4.2 C1 — Sequential Patterns (GSP / PrefixSpan, minsup = 0.5%)

Sequences were built per customer, with each customer's invoices in date order and each invoice represented as a set of product categories. PrefixSpan was run on 3,879 customer sequences with a minimum support count of 19. The most common length 2 pattern with support of 8.7% shows customers buying across all 8 categories in one session and then repeating that broad basket behaviour in their next visit. This is a strong marker of loyal, engaged shoppers. Customers who purchased only 'Other' category items and then repeated the same narrow purchase with support of 6.9% represent the opposite end, low engagement, hard to classify buyers.

Table 4: Top Sequential Patterns

Top Sequence (Simplified)	Support
All category basket → All category basket	8.74%
Narrow basket → Narrow basket	6.86%
Broad basket (no Seasonal) → Full basket	6.55%
Other only → Home Décor purchase	2.81%

4.3 C2 — At-Risk Churn Patterns

Filtering to the 3,145 At-Risk customers and re mining their sequences reveals that the dominant pattern among churned customers is repeated broad basket purchasing that eventually stops. They were fully engaged shoppers who went silent. The second most common At-Risk pattern involves customers whose last observable purchase was a narrow 'Other' category session, suggesting a gradual disengagement where they stopped exploring new categories before disappearing entirely.

4.4 C3 — Early-Warning Trigger Rules

Three trigger rules were defined for the retention team, based on the mined At-Risk sequences:

Table 5: Early-Warning Trigger Rules

Trigger Rule	Prefix Pattern	Confidence	Est. Fla
Rule 1	Customer buys across all 8 categories (broad basket)	61.7%	~83
Rule 2	Customer buys only 'Other' items (narrow basket)	42.1%	~67
Rule 3	Customer buys broad basket then drops Seasonal category	52.8%	~55

Rule 1 is the most powerful. Ironically, customers who once bought the widest range of categories are the most likely to be At-Risk, perhaps because their high engagement was occasion driven, such as gift buying season, rather than habitual. When such a customer goes 60 or more days without a new order, they should receive a targeted discount voucher. Rule 3 catches customers who are subtly narrowing their range. They stopped buying Seasonal items on their last visit, which is a measurable early warning sign 53% of the time.

5 Summary

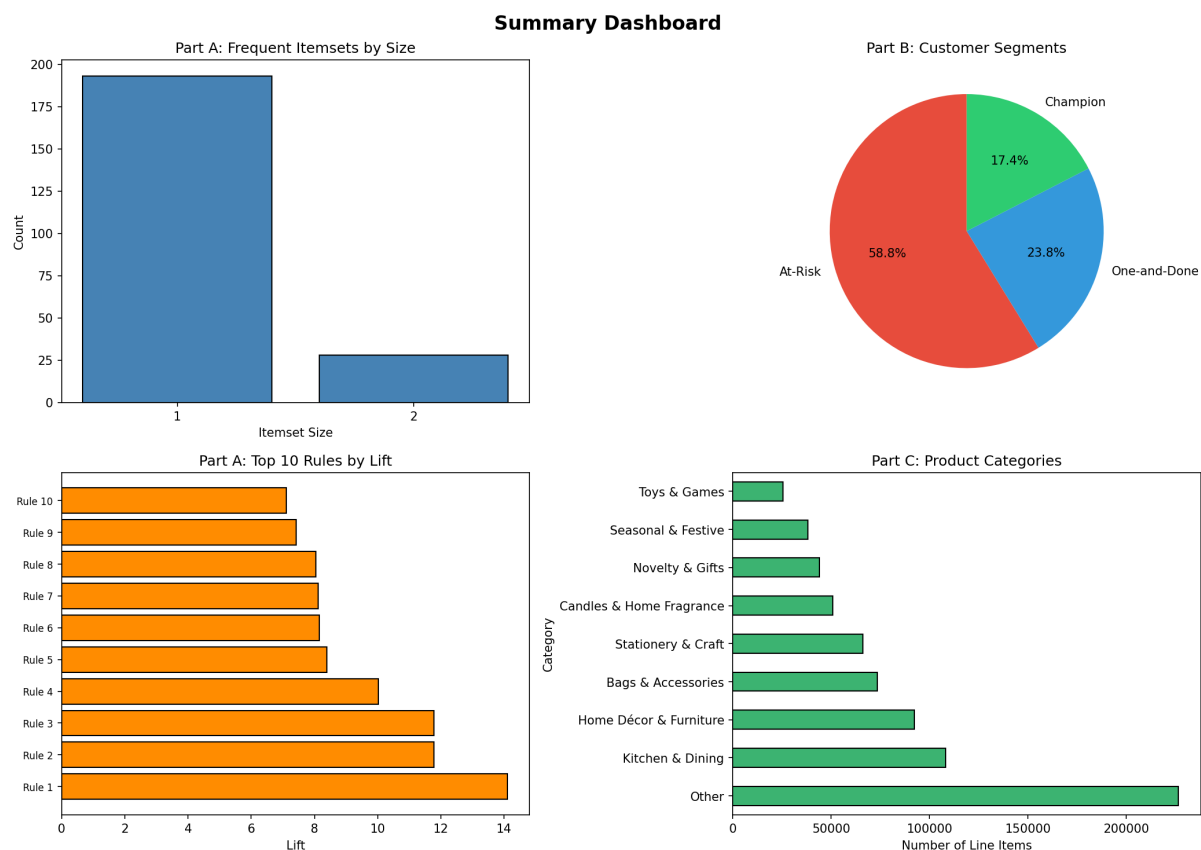


Figure 4: Summary dashboard showing itemset sizes, segment shares, top rules by lift, and category breakdown

This analysis built a full customer intelligence pipeline on two years of real UK gift retail data. FP-Growth surfaced 27 product pairs worth bundling, with ceramic trinket boxes

and wooden frames showing the strongest co purchase signals with lift of 11 to 14 times. RFM segmentation revealed that 58.8% of the customer base has gone quiet, representing a significant retention opportunity. Sequential mining confirmed that former broad basket shoppers make up the bulk of the At-Risk group, and three concrete trigger rules give the retention team an actionable early warning system to deploy targeted vouchers before churn becomes permanent.